
SCI-POINT

Engineering Conformance Specification

Scientific contract v0.1 Document revision r0.4

Concise prospective evidence procedure

Conformance status. This view defines prospective evidence against SCI-POINT-NORMATIVE-CORE/v0.1-r0.4. Typed unavailability is conforming when a required dependency is absent and the core requires fail-closed behavior. No numerical fit conforms until the compatibility method, exact numerical parent, and route binding are approved. No implementation has been assessed.

Scientific owner	Grant Wilson
Status	Scientific authority frozen; implementation conformity unassessed
Stage A packet	SCI-POINT-v0.1-r0.3-stage-b-author-packet
Stage A SHA-256	d74de333b87c66fb74c04c1346beb5ea0956bb25b61efcf9f770bb19818ee00f
Scientific directive	SCI-POINT-v0.1-r0.3-final-targeted-stage-b-owner-directive@2026-09-03
Scientific SHA-256	9f445480fd42311ebe21f4da772e4e32db400c2da08a119b699f6bd8e13a54d4
Owner directive	SCI-POINT-v0.1-r0.4-view-separation-directive@2026-09-03
Owner SHA-256	3f2331d4dc2a926ebd840cddb9bd85c9bc7d2a88be28718074313e262c259193
Normative core	SCI-POINT-NORMATIVE-CORE/v0.1-r0.4
Core SHA-256	c0ca71bd457b8e6d37a425eb3ead76400dba3a5e29c869420807928201cdcdbd
Document source	engineering-conformance.tex@v0.1-r0.4
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Build record	SCI-POINT-STAGE-B-BUILD/v0.1-r0.4@2026-09-03
Base operator	POINT-FIT/ELLIPTICAL-GAUSSIAN-COMPATIBILITY@1
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Conformance object and evidence ceiling

An implementation, representation, or product conforms only to requirements whose complete dependencies are available and whose exact identities, values or states, causes, lifecycles, and provenance are recorded. A finite number never replaces a missing authority or state. Fabricated defaults, zeros, aliases, route substitutions, source associations, covariance, response, bias, or use decisions are nonconforming.

This ECS authors no science. The normative prose, equations, typed unavailable register, edge matrix, SCI-POINT-REQ-001–038, and SCI-POINT-PRED-001–032 live only in the normative core. Evidence described here can establish contract conformance; by itself it establishes no numerical correctness, representation or response fidelity, uncertainty coverage, observational accuracy, validation, performance, readiness, or production authorization.

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1 Exact source, core, build, and route binding

The cover closes the Stage A packet, r0.3 scientific directive, r0.4 presentation directive, unchanged scientific payload, this view source, and build record. The source manifest records individual SHA-256 values. The core digest is the exact ordered concatenation of the six accepted common modules; an evidence run shall verify those bytes before applying this index.

Every numerical parent boundary record shall bind the POINT-facing role; exact upstream package product role; upstream contract version; source digest; owner approval; compatibility state; generation; and parent-response identity and status. Admissible route families are exactly MAP, JINC, FLT-FIXED, and FLT-MATCHED. Terminal FRUIT is ancestry on one such route, with terminal state and full lineage; no intermediate iteration or coadd is admissible. The evidence shall show that POINT neither selected nor substituted a route.

The parent source record also binds observation/array; estimand; signal unit, calibration, normalization, WCS/grid/frame; tangent relation; support; missing/non-finite rule; response and covari-

ance/uncertainty status; null and additive-reference state; inherited observation/exposure lineage or typed `not_applicable`; lifecycle; and provenance. POINT creates no exposure identity. Per-array products from one observation share exposure/reference terms; cross-array aggregation requires separate dependence authority.

2 Eight stage-specific logical records

Representation syntax is implementation-defined, but stages A–H shall be lossless, queryable, and distinct for each requested observation-array-parent-method atom. Co-location does not merge identities.

Stage	Record	Required logical content
A	Request and parent	Immutable request; observation/array; requested parent, method, and uses; exact parent science/identity; source reference; four applicability facts; requested search/fit domains, initialization, fallback, constraints, support/weights, and unresolved sentinels. No selected seed, fit, association outcome, realized constraint/residual, or applied/realized outcome.
B	Effective configuration	Every sentinel and requested option resolved to one explicit effective value or typed unavailable state.
C	Resolved method/plan	One frozen compatible method, route, support, objective, exact weight mapping, constraints, fallback, comparison, termination, and failure plan.
D	Application attempt	Applied plan, invocation/attempt identity, branch/restart/retry identity, execution state, cause, and provenance.
E	Realized fit/components	Selected seed; numerical values; component states; association; realized constraints; residual; diagnostics; response, bias, covariance, and uncertainty states; causes and provenance.
F	Publication candidate	Every required POINT value-or-status role, identity, cause, provenance, structural completeness, POINT publication state, and required links to requested child decisions; downstream success is not required.
G	Publication decision	POINT fit-completeness profile and bindings; decision, reasons, identity, lifecycle, and provenance.
H	Published product	Accepted immutable candidate; exact role, version, generation, inherited parent observation/exposure lineage or <code>not_applicable</code> , lifecycle, and provenance.

Producer lifecycle is the ordered vocabulary `not_requested`, `requested`, `effective`, `unavailable`, `resolved`, `applied`, `fit_realized`, `complete_publication_candidate`, `publication_decided`, `published`, `failed`, `not_produced`, and `superseded`. Component identifiability and the four SCI-VAL axes remain separate from it.

3 Availability precedence and terminal evidence

For each requested product, evaluate in this order: exact parent and route; method authority; coordinate/source boundary; applicability facts; producer attempt and realization; component identifiability; association; the role-specific dependency; then each requested named-use profile. Stop only the dependent product or claim and preserve independent sibling facts.

An early stop records only stages actually reached and an exact terminal **unavailable**, **failed**, or **not_produced** state with cause and release condition. It never fabricates **applied** or **fit_realized**, never silently deletes a requested row, and never imputes a failed array. A written unknown-scope or structural-conflict accounting artifact may be **realized**; **failed** is reserved for failure to produce the decision artifact. No defective or absent artifact asserts eligibility.

The POINT fit-completeness decision at G is the sole named-use decision in base fit completion. Pointing-support, telescope-QC, and CAL/TolProj evaluations are independently requested immutable children of F/H. Each binds the owner profile, immutable fit, SCI-VAL tuple, cause, lifecycle, action, and provenance; none mutates or regenerates the fit, rescues another use, or propagates a sibling decision. **diagnostic_display_only** is a consumer action, not an axis or producer state; **diagnostic_only** is prohibited in those roles.

4 Method-record acceptance criteria

Each authority remains **unavailable_pending_separate_owner_approval**. The following fields are stated once; all later product rows reference this section and stages C–E.

POINT-COMPATIBILITY-METHOD v0.1. One immutable, content-bound, owner-approved record shall specify source equation and parameter ordering; width/angle conventions; domains, constraints and boundaries; rows, residual and objective; exact consumed W_{fit} and every source mapping’s identity, units, normalization, support, lifecycle, ceiling and provenance; expected-position/ search distinction; search, score, candidates, interpolation, ties, initialization, fallback, association, retry and sentinels; solver-independent comparison; convergence, rank, identifiability, failure and compatibility; and applicability tests with typed unavailability.

POINT-FORMAL-ERROR-METHOD v0.1. Its separate record shall specify fitted parameterization; stochastic model and weight relation; exact covariance/error construction; curvature, residual scaling, degrees of freedom, rank and positive-definiteness; transformations and marginal extraction; bounds, circularity, degeneracy and weak identification; consistency, units, scope, omissions and failure states; signed standardization; and exact alias approval.

POINT-FULL-MAP-RMS-METHOD v0.1. Its separate record shall specify exact signal parent and generation; population, centering/mean, weighting, source-region treatment, support, edge and validity; non-finite and zero-denominator behavior; units, lifecycle and provenance; relation to the amplitude parent; signed ratio; and exact alias approval.

No method is released by a familiar name or presumed default. The compatibility record gates every numerical fit. The formal-error record gates marginal errors and formal standardization but not otherwise authorized fitted values. The full-map-RMS record gates only its RMS scale, signed diagnostic, and any explicitly approved alias.

5 Product and companion record obligations

The realized-fit record at E shall bind selected branch and seed; association method, domain, state, and cause for every branch; support and row count; objective and exact W_{fit} mapping; six role values, units, and conventions with per-component states; constraints and rank; termination, convergence, and retry; model or reconstructibility; residual; diagnostics; parent null/additive state; all response, bias, covariance and uncertainty identities/states; and every cause and provenance link. Numerical A has the exact fit-amplitude-component role bound in the core; only established association creates the separately typed source-associated-amplitude role.

The displacement product at F/H shall reference E and bind expected-position identity; fitted centroid; ordered basis, handedness and positive directions; arcsecond unit; coordinate state/time; transform/Jacobian; source and coordinate uncertainty states; the signed equation $\Delta_{\text{POINT}} = \hat{\mu} - \mu_{\text{expected}}$; support, validity, lifecycle and provenance. It carries no correction sign or application claim.

Each response companion at E/F shall identify exactly one scope: fixed POINT, fixed parent, POINT-full-procedure/parent-fixed, whole-chain full procedure, or empirical bias/accuracy. It binds comparable numerical baseline/perturbed products and states; source and association; parameter roles and gauge/order; units/basis and component availability; perturbation source/domain/direction/ unit/ ϵ and domain relation; branch changes; finite/analytic convention; method; cause and provenance. Incomparable outputs are unavailable or typed discontinuities/state transitions, never differences of lifecycle objects. Whole-chain evidence requires complete upstream rerun authority; finite differences do not compose as Jacobians without an exact theorem.

Conditional displacement covariance at E/F binds every variance and cross-covariance in $C_{\Delta} = C_{\hat{\mu}} + C_{\mu_{\text{expected}}} - C_{\hat{\mu}, \mu_{\text{expected}}} - C_{\mu_{\text{expected}}, \hat{\mu}}$, the common basis, transforms and Jacobians, independence authority for any zero cross term, validity, state, cause and provenance. Missing terms make the combined covariance unavailable, not zero or diagonal.

6 Requirement-to-evidence index

The normative-core section keys are: N, notation and state axes; O, operation, ownership, and routes; M, model, measurement, and equations; G, inputs, gates, and dependencies; and V, validity, unavailable states, and edge rules. Each REQ-**nnn** below denotes exact SCI-POINT-REQ-**nnn**; the core prose governs.

Req.	Core	Required evidence object	Comparison class	Terminal result
REQ-001	O,G	A–H atom and G decision	structural cardinality	unavailable / failed / not produced
REQ-002	G	A–H field and lineage record	exact identity/presence	unavailable
REQ-003	O	A–C parent-role record	closed enumeration	unavailable
REQ-004	O	C route-selection trace	forbidden transition	unavailable
REQ-005	O	A–C FRUIT ancestry closure	terminal identity	unavailable

Req.	Core	Required evidence object	Comparison class	Terminal result
REQ-006	O,G	A/E parent-response and mismatch record	identity/status	unavailable
REQ-007	M,G	A/E four-object coordinate record	distinct identity	unavailable
REQ-008	M	A/E/H tangent-boundary record	geometric transform	unavailable / failed
REQ-009	M	E/F/H displacement product	signed equation/unit	unavailable
REQ-010	M	C/E model reconstruction	equation/family	failed / nonconforming
REQ-011	M	C/E shape and gauge record	invariant and symmetry	undefined / unavailable
REQ-012	G	C-E method and route approvals	authority presence	unavailable
REQ-013	M,G	C/E exact weight mapping	typed identity/content	unavailable
REQ-014	M,G	C-E lifecycle and branch trace	ordered state sequence	unavailable / failed / not produced
REQ-015	M	E association per branch	independent gate	unavailable / failed
REQ-016	O,G	A/B applicability facts	typed authority/domain	unavailable
REQ-017	M,G	D/E fit and residual record	reconstruction	unavailable / failed
REQ-018	N,G	E/F component-state vector	closed enum/cardinality	unavailable / failed
REQ-019	O,M	E/F amplitude roles and association	role/dependency	unavailable
REQ-020	O,M	E/F shape-role publication	claim ceiling	unavailable
REQ-021	G	E/F formal-error record	method authority	unavailable
REQ-022	G	E/F covariance status	absence semantics	unavailable
REQ-023	M,G	E/F derived-product record	equation, identity, gates	unavailable
REQ-024	M,G	C/F alias/admission record	explicit authority or absence	unavailable
REQ-025	M,G	E response-family record	scope and comparison	unavailable / discontinuity
REQ-026	G	F role-dependency evaluation	product-local blocking	unavailable
REQ-027	N	A/G SCI-VAL record	exact axes/crosswalk	incomplete / failed / not produced
REQ-028	N	G action and axis fields	token-role separation	unavailable / failed
REQ-029	O	G owner/profile record	ownership identity	unavailable
REQ-030	N,O	G draft-profile binding	registration and digest	unavailable

Req.	Core	Required evidence object	Comparison class	Terminal result
REQ-031	O	F–H output-role inventory	prohibited-product absence	not produced
REQ-032	O	E–H sibling-array rows	atomic preservation	failed / not produced
REQ-033	M	E residual/additive record	reconstruction and state	unavailable / failed
REQ-034	N,V	A–H terminal/edge evidence	exact enum/cause	unavailable / failed / not produced
REQ-035	M,G	E/F covariance terms and transforms	exact matrix/basis	unavailable
REQ-036	M,G	E rerun/response authority trace	scope and comparison	unavailable / discontinuity
REQ-037	N,O	G child-decision record	exact cross-walk/lineage	incomplete / failed / not produced
REQ-038	G	Claim-and-evidence ledger	ceiling/absence	no higher claim

7 Prediction-to-fixture index

Each PRED-**nnn** below denotes exact SCI-POINT-PRED-**nnn**. A future test is admissible only after its method, route, and oracle dependencies are approved; the prediction itself is not a claim that testing occurred.

Prediction ID	Fixture class	Oracle/reference	Comparison	Governing requirements
PRED-001	Noiseless matching Gaussian	Approved method solution	Six roles, circular/bound cases	010–014,018
PRED-002	Authorized signal scaling	Method amplitude law	Amplitude scaling; fixed reference	006,007,019
PRED-003	Signed offset	Exact expected position	Centroid minus expected; no correction	008,009,031
PRED-004	Search-center perturbation	Fixed source reference	Branch may change; zero does not	007,009,014
PRED-005	WCS/basis transform	AST transform oracle	Physical components/handedness	008,009
PRED-006	Central/fallback branches	Stage C–E records	Branch distinguishability	001,014
PRED-007	Exact score tie	Approved tie rule	Selected state only after D	002,012,014
PRED-008	Wrong global peak	Association oracle	Fit amplitude survives; source roles blocked	015,019
PRED-009	Principal-axis exchange	Gaussian invariance	Model equality after angle transform	011
PRED-010	Exact circularity	Symmetry oracle	Equal widths; orientation undefined	011,018

Prediction ID	Fixture class	Oracle/reference	Comparison	Governing requirements
PRED-011	Active parameter bounds	Applied/realized constraints	Bound-censored component state	014,018
PRED-012	Constant/gradient mismatch	Known injected structure	Bias/residual visibility	017,033
PRED-013	Missing/non-finite support	Approved support rule	Reached stages and terminal state only	002,014,034
PRED-014	Reliability weights	Weight-authority record	No inverse-covariance alias	013
PRED-015	Degenerate/error cases	Formal-method oracle	Values may survive; errors unavailable	018,021,034
PRED-016	Route-response mismatch	Exact parent response record	Compatibility/bias state retained	003,006,025
PRED-017	Asymmetric parent response	Bias/response authority	Typed bias; independent fit survives	006,025,026
PRED-018	FRUIT terminal cases	Ancestry closure	Terminal admitted; intermediate rejected	005
PRED-019	Array or component failure	Per-array atom inventory	Siblings preserved; no global success	001, 018, 031, 032
PRED-020	Arbitrary input	Output-role inventory	No POINT aggregate or correction	031
PRED-021	Signed amplitude-to-RMS	RMS-method oracle	Sign retained; bad denominator unavailable	019, 023, 034
PRED-022	Signed formal ratio cases	Formal-method oracle	$\hat{A}/\sigma_A$; magnitude distinct	019, 021, 023, 024
PRED-023	One fit, multiple uses	Owner profile records	Independent SCI-VAL tuples and actions	027, 028, 037
PRED-024	Missing or conflicting profile	Bound SCI-VAL crosswalk	Realized accounting; failure only on production	027, 030, 034, 037
PRED-025	Three response scopes	Rerun oracle by scope	No alias; typed incomparability	025,036
PRED-026	Missing compatibility method	Authority registry	No numerical fit/product	012
PRED-027	Missing formal-error method	Authority registry	Values survive; formal products unavailable	021,022,026
PRED-028	Missing full-map-RMS method	Authority registry	Only RMS-dependent roles blocked	019,023,026
PRED-029	Every search/retry branch	Association oracle	Source roles require established association	015,019,023

Prediction ID	Fixture class	Oracle/reference	Comparison	Governing requirements
PRED-030	Missing response companions	Separate authority registries	Separate release; unrelated roles survive	006, 025, 026, 035, 036
PRED-031	Child failure or absence	Immutable parent-child lineage	No rescue, mutation, generation, or base block	001, 028, 037
PRED-032	Shared reference error	Full joint covariance oracle	Cross terms retained; no false averaging	035

8 Parity, source closure, and nonclaims

The r0.4 parity record shall verify: unchanged six-file scientific bytes; gap-free REQ-001–038, PRED-001–032, and UNAV-001–023 identities; equation, product-role, lifecycle, component-state, SCI-VAL, response-family, diagnostic, weight, exposure-lineage, owner-decision, source, and build parity; and exact agreement between this index and the normative core. The rationale is an explanatory view; this ECS is the evidence procedure; neither overrides or extends the core.

The package has one-to-one coverage of ODQ-001 through ODQ-009, including ODQ-003A/003B. The remaining actions are approval of the three exact methods, exact route instances, four draft profiles, and any separately requested response, bias/accuracy, or covariance authority. The deterministic delivery packet contains the Stage A archive and sidecars, both owner directives, the unchanged common source, all three document sources and PDFs, manifests, parity/semantic reports, build recipe, verifier, and all-page QA record.

This specification claims no numerical route, implementation inspection or conformity, representation or response/covariance fidelity, uncertainty coverage, observational pointing accuracy, validation, performance, readiness, production suitability, production authorization, or Unity activity.